

Full-Database Heterogeneity Validation

Sphere Region Method, Pflugfelder HI & WEPL ($n = 69\,295$, $r = 15\text{ mm}$)

Mikołaj Stryja

Department of Radiation Science & Technology

24 April 2026

Outline

- 1 Recap & New Approach
- 2 Sphere Region Method
- 3 New Metrics: Pflugfelder HI & WEPL
- 4 Results: Full Database
- 5 Energy-Stratified Analysis
- 6 Case Studies
- 7 Sphere Radius Sensitivity
- 8 Discussion & Key Findings
- 9 Next Steps

Recap & New Approach

Last week ($n = 3\,950$): Spherical neighbourhood around BP $\rightarrow$ six advanced HU metrics + 3-D Sobel edge metrics, correlated against GPR. Best predictor: $S_{95, \text{BP}}$ (Spearman $\rho = -0.391$).

This week: three major developments

- 1 **Full-database scale-up**: run on the complete ADoTA training set ($n = 69\,295$ beamlets, $18\times$ larger).
- 2 **Sphere region method**: replace the variable-length BP-range extraction with a *fixed* sphere of radius r centred on the Bragg-peak voxel. Two radii tested: $r = 15$ mm (complete) and $r = 5$ mm (in progress).
- 3 **New metrics**: Pflugfelder heterogeneity index (HI) and Water-Equivalent Path Length (WEPL) mean & std.

Dataset: $n = 69\,295$ beamlets from the ADoTA training set, beam energies 70 MeV to 250 MeV. Model: DoTA_v3_grid_search_v11.

Methodology Change: Sphere Region

Previous (BP-range method):

- Proximal boundary $z_{\min}$: walk backward from BP to where $\text{IDD} < 0.50 \text{IDD}_{\max}$.
- Distal boundary $z_{\max}$: walk forward to $\text{IDD} < 0.10 \text{IDD}_{\max}$.
- Region spans all slices in $[z_{\min}, z_{\max}]$.
- *Variable length*: shallow beams ≈ 10 mm, deep beams ≈ 50 mm.

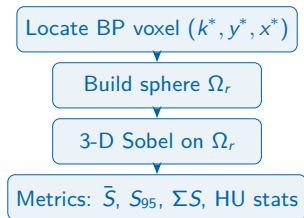
New (sphere method):

- Locate BP voxel (k^*, y^*, x^*) from $\arg \max_k \text{IDD}[k]$ and lateral dose peak.
- Include all voxels within distance r of BP:

$$\Omega_r = \{(k, y, x) \mid \|\mathbf{p} - \mathbf{p}^*\|_{\text{mm}} \leq r\}$$

where $\mathbf{p} = (k\Delta z, y\Delta y, x\Delta x)$.

- *Fixed geometry*: radius r is a single hyperparameter; independent of beam energy.



	BP-range	Sphere
Geometry	variable	fixed
Length	$\propto E$	$r = \text{const}$
Hyperpar.	α, β fractions	r [mm]
Lateral	full beam	within r

New Metrics: Pflugfelder HI & WEPL

Water-Equivalent Path Length (WEPL)

For each lateral ray (y, x) through the CT:

$$\text{WEPL}(y, x) = \Delta z \sum_{k=0}^{k^*} \text{RSP}(k, y, x)$$

RSP = relative stopping power (HU $\rightarrow$ RSP via Schneider calibration); k^* = BP slice index.

WEPL map $\in \mathbb{R}^{H \times W}$: water-equivalent range to BP per lateral ray.

Output scalars per beamlet:

- `wep1_mean` = $\mu(\text{WEPL})$ [mm]
- `wep1_std` = $\sigma(\text{WEPL})$ [mm]
- `pflugfelder_hi` = HI

Heterogeneity Index (HI) [1]:

$$\text{HI} = \frac{\sigma(\text{WEPL})}{\mu(\text{WEPL})}$$

Computed over the flux footprint (10 % threshold).

Interpretation:

- HI ≈ 0 : homogeneous path (pure water).
- HI $\gg 0$: large lateral stopping-power variation $\rightarrow$ range uncertainty.

Motivation: operates in dose space (mm of water) rather than image space (HU) — complementary to Sobel and HU metrics.

Pflugfelder, Wilkens, Oelfke, *Phys. Med. Biol.* 53(6):1689, 2007.

Dataset & Model Inference ($n = 69\,295$)

Dataset:

- $n = 69\,295$ beamlets (full ADoTA training set)
- 853 skipped (zero flux or duplicate)
- Energy range: 70 MeV to 250 MeV
- Grid: $160 \times 30 \times 30$ voxels, $2 \times 2 \times 2$ mm

Model: ADoTA v3 (DoTA_v3_grid_search_v11)

Gamma: 2%/2 mm, 10 % lower dose cutoff

GPR distribution (overall):

Mean = 98.83%, $\sigma = 1.47\%$

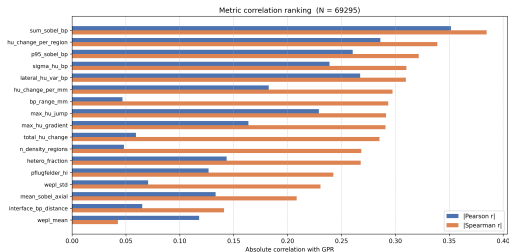
Range: [57.22%, 100.00%]

Energy bin distribution:

Bin (MeV)	n	GPR (%)
70–100	16 163	98.46 ± 1.85
100–130	15 869	98.76 ± 1.58
130–160	16 995	98.91 ± 1.41
160–190	13 881	99.21 ± 0.96
190–220	3 231	99.16 ± 0.63
220–250	3 156	98.63 ± 0.76

Pattern from last week confirmed at $18\times$ scale: lower energies show larger GPR variance.

Overall Correlation Ranking ($n = 69\,295$, $r = 15\text{ mm}$)

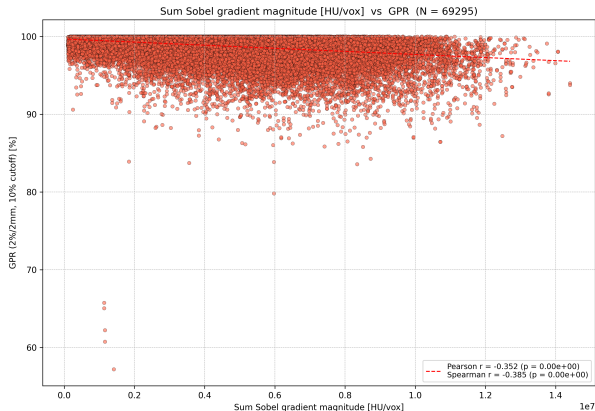


Metric	Spearman	Pearson
sum_sobel_bp	−0.385	−0.352
hu_change_per_region	−0.339	−0.286
p95_sobel_bp	−0.322	−0.260
σ_{HU} at BP	−0.310	−0.239
lateral_hu_var_bp	−0.310	−0.267
hu_change_per_mm	−0.297	−0.183
bp_range_mm	−0.293	−0.047
max_hu_jump	−0.291	−0.229
max_hu_gradient	−0.291	−0.164
total_hu_change	−0.285	−0.059
n_{regions}	−0.268	−0.048
hetero_fraction	−0.268	−0.143
pflugfelder_hi	−0.242	−0.127
interface_bp_dist.	+0.141	+0.065

All $p \approx 0$. **Bold:** new metric. Spearman $\gg$ Pearson confirms monotonic but non-linear relationships.

Key shift: ΣS_{BP} (sum Sobel) now leads instead of $S_{95, \text{BP}}$ — Sobel aggregation changes with sphere geometry.

Best Predictor: Sum Sobel in BP Sphere



ΣS_{BP} achieves Spearman $\rho = -0.385$ (Pearson $r = -0.352$, $p \approx 0$, $n = 69\,295$). Compare with last week: $S_{95,BP}$ led with $\rho = -0.391$ (BP-range, $n = 3\,950$). Results are consistent across methods and at $18\times$ larger scale.

Motivation for Energy Stratification

Observation (confirmed at full scale): lower-energy bins show greater GPR variance ($\sigma \approx 1.85\%$ at 70–100 MeV vs. 0.63% at 190–220 MeV).

Question: are heterogeneity–GPR correlations genuine, or are they confounded by beam energy?

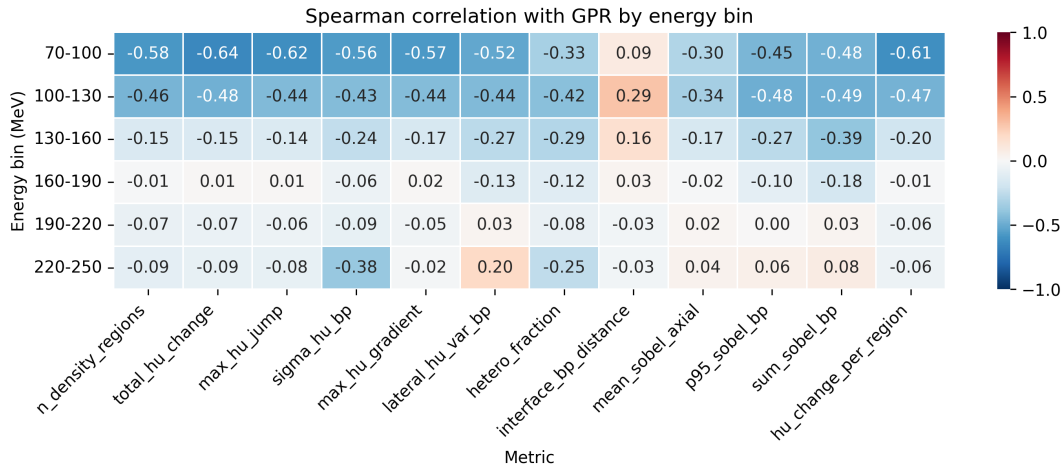
Three complementary analyses (same as last week, now at scale):

- 1 **Binned correlations:** Spearman ρ within each 30 MeV energy bin independently.
- 2 **Energy-coloured scatter plots:** visualise metric vs. GPR coloured by energy.
- 3 **Partial correlations:** partial Spearman controlling for energy,

$$\rho_{XY.Z} = \frac{\rho_{XY} - \rho_{XZ} \rho_{YZ}}{\sqrt{(1 - \rho_{XZ}^2)(1 - \rho_{YZ}^2)}}$$

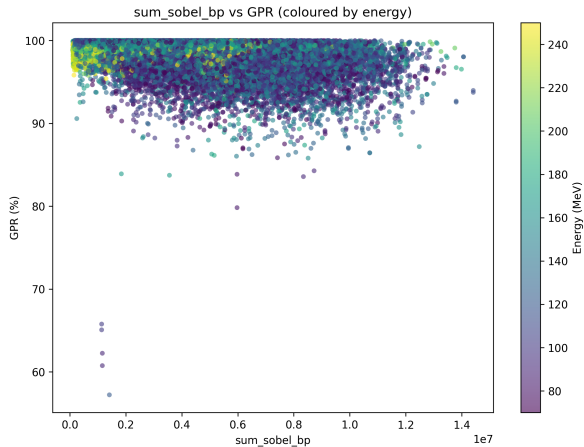
where X = metric, Y = GPR, Z = energy.

Spearman Correlation by Energy Bin



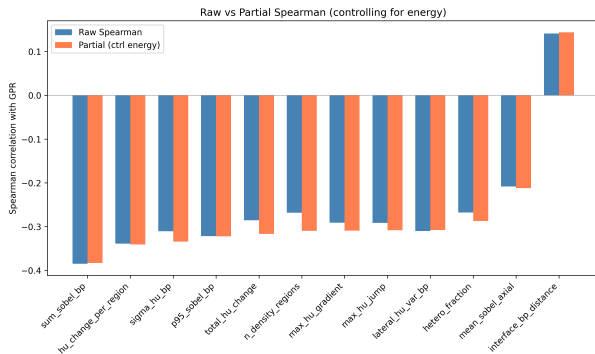
Correlations are **strongest at low energies**. 70–100 MeV: $|\rho| > 0.60$ for HU-change metrics and > 0.48 for Sobel metrics. Above 160 MeV, correlations largely vanish; at 190–220 MeV, $\rho \approx 0$ for most metrics.

Energy-Coloured Scatter: Sum Sobel at BP



Low-energy beamlets (dark) cluster in the high- ΣS / low-GPR region; high-energy beamlets (bright) populate the compressed-metric / high-GPR region. The energy gradient is visible but does *not* drive the correlation (see partial analysis).

Partial Correlations Controlling for Energy

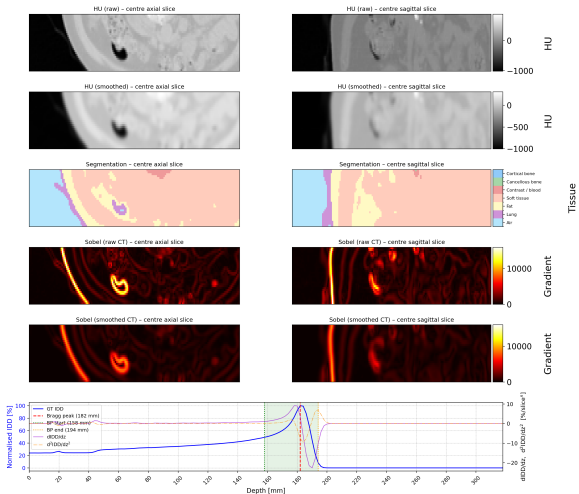


Metric	Raw	Partial
sum_sobel_bp	−0.385	−0.383
hu_change/region	−0.339	−0.341
$\sigma_{\text{HU,BP}}$	−0.310	−0.334
p95_sobel_bp	−0.322	−0.322
n_{regions}	−0.268	−0.309
max_hu_gradient	−0.291	−0.309

Raw and partial correlations differ by $|\Delta| < 0.04$: the heterogeneity–GPR association is **not confounded by energy**. For n_{regions} ($\Delta = -0.041$), controlling for energy *strengthens* the correlation, mirroring last week's finding.

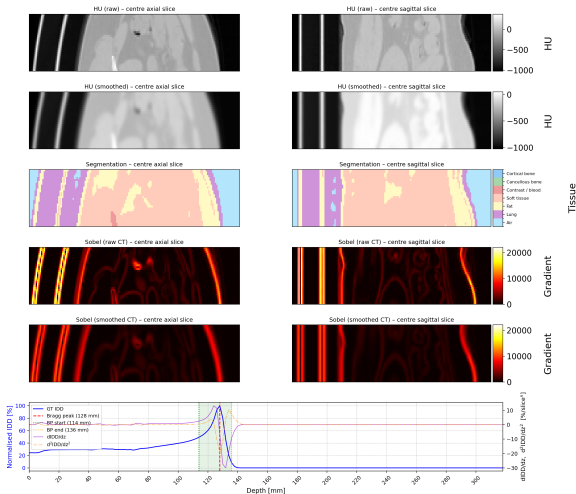
Worst-GPR Beamlet: 1f279484, 120.67 MeV, GPR = 57.22 %

CT HU & Segmentation - 1f279484-7da5-4193-9fcb-0ac81c6bf1b6



Best-GPR Beamlet: ffa7d740, 93.87 MeV, GPR=100.00 %

CT HU & Segmentation - ffa7d740-9119-4f00-ad47-e818e0584544



Sphere Radius: 15 mm vs 5 mm ($n = 69\,295$)

Metric	$\rho_s^{15\text{mm}}$	$\rho_s^{5\text{mm}}$	$\Delta \rho $
sum_sobel_bp	−0.385	−0.295	+0.090
p95_sobel_bp	−0.322	−0.292	+0.030
mean_sobel_axial	−0.209	−0.202	+0.007
hu_change_per_region	−0.339	−0.339	0.000
σ_{HU} at BP	−0.310	−0.310	0.000
lateral_hu_var_bp	−0.310	−0.310	0.000
hu_change_per_mm	−0.297	−0.297	0.000
max_hu_jump	−0.291	−0.291	0.000
max_hu_gradient	−0.291	−0.291	0.000
total_hu_change	−0.285	−0.285	0.000
n_{regions}	−0.268	−0.268	0.000
hetero_fraction	−0.268	−0.268	0.000
pflugfelder_hi	−0.242	−0.242	0.000
wepl_std	−0.231	−0.231	0.000
interface_bp_dist.	+0.141	+0.141	0.000

Key observation: *only the Sobel metrics depend on sphere radius.* All HU and WEPL/Pflugfelder metrics are radius-invariant.

Why? The HU metrics ($\text{hu_change_per_region}$, $\sigma_{\text{HU,BP}}$, ΔH_{max} , ...) are computed on the *flux-weighted axial HU profile* over $[z_{\text{min}}, z_{\text{max}}]$ and on the *lateral BP slice*. WEPL is computed over the full lateral footprint up to k^* . Only the Sobel-family aggregation is bounded by the sphere Ω_r .

Effect on Sobel metrics:

- ΣS_{BP} : $|\rho_s|$ drops from 0.385 $\rightarrow$ 0.295 (−23% relative)
- $S_{95,\text{BP}}$: 0.322 $\rightarrow$ 0.292 (−9%)
- $\bar{S}_{\text{axial}}$: essentially unchanged

Recommendation: adopt $r = 15$ mm as the default — strictly dominates $r = 5$ mm on

Key Findings

- 1 **Scale confirmed:** findings from last week ($n = 3\,950$) fully replicated at $n = 69\,295$ ($18\times$ larger). Sobel-based metrics remain the strongest single predictors of GPR degradation.
- 2 **Sphere method:** switching from BP-range to sphere changes the leading metric from $S_{95,BP}$ to ΣS_{BP} (sum vs. percentile), but both reach $|\rho_s| \approx 0.38\text{--}0.39$ — essentially equivalent predictive power.
- 3 **Energy-stratification at scale:** correlations strongest at 70–100 MeV ($|\rho_s| > 0.60$ for HU-change metrics), weaken above 160 MeV. Partial correlations confirm energy is *not* a confounder ($|\Delta\rho| < 0.04$).
- 4 **Pflugfelder HI:** achieves Spearman $\rho = -0.242$ (rank #13 of 17). Clinically interpretable (water-equivalent units) but weaker than Sobel metrics. WEPL std ($\rho = -0.231$) is similar.
- 5 **Radius sensitivity:** only Sobel metrics depend on the sphere radius. ΣS_{BP} drops from $\rho_s = -0.385$ at $r = 15$ mm to -0.295 at $r = 5$ mm (a 23% relative loss); HU and WEPL/Pflugfelder metrics are unchanged. **Recommendation:** use $r = 15$ mm.

- 1 **Interface Severity Index (ISI)** (in development): per-class interface-pair severity matrix ($W_{ab} = (\Delta RSP)^2$) using the Schneider 24-class decomposition, aggregated within the $r = 15$ mm BP sphere. First correlations expected next week.
- 2 **Composite difficulty score**: combine top metrics (ΣS_{BP} , $\sigma_{HU,BP}$, $\Delta H/n_{reg}$) into a single “difficulty index” via PCA or logistic regression on the full $n = 69\,295$ dataset.
- 3 **VLM difficulty estimation**: implement Track A (zero-shot) as a proof-of-concept and compare Spearman ρ against engineered metrics on a held-out subset.
- 4 **Worst-case deep-dive**: visualise and analyse the beamlets with GPR $< 90\%$ in the 70–100 MeV bin to identify anatomical patterns driving failure.

Questions?



D. Pflugfelder, J. J. Wilkens, U. Oelfke, "Worst case optimization: a method to account for uncertainties in the optimization of intensity modulated proton therapy," *Phys. Med. Biol.*, vol. 53, no. 6, pp. 1689–1700, 2007.

<https://doi.org/10.1088/0031-9155/53/6/013>